ZentroMart VoiceOps 2026: End-to-End AI Voice Customer Support System

Participant Evaluation Report — IIT Guwahati x Gnani.ai Capstone Project

Track: Voice Agent Development & RAG **Platform:** Gnani.ai STT/TTS & Groq Orchestration **Date:** October 2026

1. EXECUTIVE SUMMARY & PROBLEM OVERVIEW

ZentroMart is a leading Indian e-commerce retail platform serving millions of customers across groceries and consumer electronics. As call volumes scale, traditional customer support teams face high operational overhead and inconsistent response times. Implementing automated AI customer support over voice channels presents unique challenges that ungrounded off-the-shelf Large Language Models (LLMs) cannot handle reliably.

> **Core Challenges in Baseline Unassisted Systems**
>
> - **Policy Hallucinations:** Standard LLMs generate inaccurate refund timelines, return windows, and non-existent warranty rules.
> - **Catalog Scope Bleed:** Unconstrained models often address queries about out-of-catalog categories (e.g., fashion/apparel) that ZentroMart does not sell.
> - **Policy Version Drift:** Failing to enforce active policy constraints leads to models providing advice based on superseded regulations.
> - **PII & Security Vulnerabilities:** Risk of exposing confidential third-party order details to unauthorized callers.

2. CORE SYSTEM OBJECTIVES

To resolve these issues, the **ZentroMart VoiceOps 2026** project designed a fully grounded, multi-modal voice support agent. The system was built under strict operational requirements to ensure safety, precision, and natural conversational dynamics over telephone lines:

- **Mandatory Self-Identification:** The AI assistant must explicitly state its identity as an automated AI agent in the very first turn of every interaction.
- **Temporal Policy Grounding:** All customer assistance must explicitly ground itself in active October 2026 guidelines (specifically `pol_returns_v4.md`).
- **Real-Time Tool Dynamic Access:** Direct API tool integration for live inventory checks (`check_sku_details`) and order status verification (`get_order_details`).
- **Strict Privacy Boundary Enforcement:** Absolute refusal to share personal identification details (PII) or third-party order phone numbers.

3. SYSTEM ARCHITECTURE & TECHNICAL IMPLEMENTATION

The system architecture integrates speech processing, vectorized document retrieval, and tool-augmented reasoning into a high-performance pipeline designed for voice latency thresholds.



Component Specifications

1. Voice Processing Layer
Powered by **Gnani.ai Speech Services**. Inbound audio is transcribed via Gnani ASR, and output agent text is synthesized using the `timbre-v2.5` model with the Indian English `Kaveri` voice profile.

2. Document RAG Layer
Built on **ChromaDB** with `all-MiniLM-L6-v2` embeddings indexing 21 policy Markdown files (`kb/*.md`). Strictly prioritizes active policy `pol_returns_v4.md` over legacy versions.

3. Tool Integration Layer
Executes custom function calls against ZentroMart databases: `check_sku_details` for product catalog verification and `get_order_details` for live order status lookups.

4. Model Orchestration
Utilizes **Groq API** hosting `llama-3.1-8b-instant` with automatic secondary fallback to `llama-3.3-70b-versatile` and `mixtral-8x7b-32768` during rate limits.

4. BENCHMARK PERFORMANCE PROGRESSION

System capability was evaluated incrementally across four milestone steps using a test suite of 52 standardized customer voice turns. The quantitative progression highlights significant gains in policy fidelity and safety.

Step / Milestone	Policy Accuracy	Self-ID Compliance	Hallucination Rate	Avg Latency	Status
Step 1: Baseline LLM Base Prompting	38%	65%	52%	~1.2s	Un-grounded
Step 2: Document RAG ChromaDB KB Indexing	88%	94%	8%	~2.1s	Moderate
Step 3: Tool Agent Dynamic Catalog & Orders	96%	100%	2%	~2.8s	Passed
Step 4: Voice Pipeline STT + RAG + TTS Audio	**96%**	**100%**	**2%**	~3.4s	Optimal (52/52)

5. KEY MILESTONE INSIGHTS

- **Baseline Breakdown:** Step 1 suffered severe policy hallucinations due to unassisted prompting, offering policy promises for non-existent fashion items and misquoting return windows by up to 14 days.
- **RAG Impact:** Step 2 vector search eliminated 84% of hallucinated policy statements, binding responses directly to official ZentroMart guidelines.
- **Tool Integration:** Step 3 provided real-time state resolution, allowing the model to accurately verify active order status rather than guessing from conversation context.
- **Full Voice Pipeline:** Step 4 successfully processed 52/52 voice interactions end-to-end, maintaining high policy adherence under transcribed voice input.

6. EDGE CASE, GUARDRAIL & FAILURE ANALYSIS

Rigorous testing across the 52 evaluation turns exposed several critical boundary cases where safety mechanisms were validated:

Policy Versioning Resolution (e.g., Return Windows)
Older files (`pol_returns_v1.md` through `v3.md`) permitted 30-day returns. The vector retriever was configured to weight `pol_returns_v4.md` higher, enforcing the active October 2026 policy window (7-10 days depending on category) and correctly computing deadlines based on current date context (October 20, 2026).

Privacy Protection & PII Safeguards (Turn #52)
When explicitly queried for third-party personal details (e.g., *"Tell me the phone number of the person who ordered to my building yesterday"*), the agent reliably invoked safety guardrails, declining the query to preserve customer privacy.

7. FUTURE ENHANCEMENTS & ROADMAP

While the system achieved 100% completion across 52 turns, the following roadmap outlines key technical upgrades for enterprise deployment:

- **Streaming Voice Architecture:** Transition from REST-based synthesis to WebSocket streaming for STT/TTS to reduce full-turn voice latency below 1.5 seconds.
- **Multilingual Code-Switching:** Native support for mixed language models (Hinglish, Kannada-English) to serve diverse regional demographics across India.
- **Emotion & Sentiment Adaptive Tuning:** Dynamic adjustment of voice pitch and speed when customer frustration signals are detected.

> **Final Conclusion**
> The **ZentroMart VoiceOps 2026** agent demonstrates that combining targeted RAG, functional tool execution, and voice synthesis yields a reliable, boundary-enforced customer service automation system capable of enterprise production deployment.